

ISHNA TRIPATHI



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[Ishna Tripathi](#)



[Github Profile](#)

PROFILE

Physics and Astrophysics undergraduate at Trinity College Dublin with a strong foundation in, machine learning, data analysis and statistical modelling. Experienced in applying probabilistic machine learning techniques, including Gaussian Process regression, especially Python packages such as Scikit-learn, Dynesty, XGBoost, Pandas, Matplotlib, Numpy and many more to help analyse complex datasets, identify patterns in noisy data and quantify predictive uncertainty. Proficient in Python and scientific computing, with experience developing data analysis pipelines, simulation-based models and working with HPC environments. Combines strong quantitative problem-solving skills with leadership, stakeholder engagement and communication experience, seeking to apply technical and analytical expertise to solve complex business and financial problems.

RELEVANT PROFESSIONAL EXPERIENCE

Prof Johanna Vos, School of Physics - Trinity College Dublin

May 2025 – July 2026

Research Intern

- Investigated potential exomoon signatures around the brown dwarf 2MASS J1047 using photometric light-curve and spectral data analysis, applying Gaussian Process regression (ML) to model variability and test transit hypotheses.
- Designed and executed synthetic exomoon injection-recovery experiments to quantify detection limits, assess sensitivity, and validate candidate signals.
- Develop Python-based data analysis pipelines for processing, modelling and visualising scientific datasets.
- Work with remote High-Performance Computing environments and communicate complex technical findings through research presentations and scientific writing.

The Kaipātiki Project, Auckland, New Zealand

May – June 2026

Community Engagement & Sustainability Project Lead

- Analysed engagement and participant feedback data to evaluate community behaviour-change initiatives and identify opportunities for improvement.
- Led the "10 by 10" litter-prevention campaign, achieving a 55% conversion rate and engaging 133+ residents through targeted outreach and stakeholder engagement.
- Developed data-informed recommendations, educational resources, and sustainability programmes, collaborating with community groups to deliver events including a 600-native-plant giveaway.

Prof Sumanta Sarkhel, Indian Institute of Technology Roorkee

July – August 2025

Research Intern

- Designed and built a drone-mounted solar irradiance and UV monitoring system, integrating photodiodes, UV sensors, and environmental measurement modules.
- Developed Arduino-based data acquisition and control software while performing circuit design, soldering, sensor calibration, and hardware optimisation for aerial deployment.
- Applied experimental physics, embedded systems, and instrumentation techniques to support accurate atmospheric and environmental data collection.

Other independent projects on available on [GitHub](#).

EDUCATION

Trinity College Dublin

2023-2027

B.A. (Hons) Physics and Astrophysics

Castleknock Community College

2020-2023

Leaving Certificate: 566 points

CERTIFICATION

Oxford Character Project

- Engaged in workshops and discussions with peers and mentors, to explore frameworks for resilience, integrity, empathy, and purpose in professional and academic contexts.
- Worked on character development, ethical leadership, and values-driven decision making.

ACHIEVEMENTS

Laidlaw Scholar

- Attained the prestigious Laidlaw Scholars Leadership and Research Programme, designed to develop passionate, ethical, and impact-driven leaders.
- Selected as a Laidlaw Scholar in recognition of leadership potential and academic promise.

TEDxTrinityCollegeDublin

- Delivered a TEDx talk titled Cloudy with a Chance of Exomoons – Hunting for Transiting Exomoons.

PUBLICATIONS

Taylor and Francis

Cloudy with a Chance of Exomoons: Hunting for Transiting Exomoons Around a Brown Dwarf

SKILLS

Programming & Computing: Python – Scikit-learn, XGBoost, Pandas, NumPy, Arduino, LaTeX, HTML, CSS, VS Code, remote HPC environments

Data & AI Foundations: Data analysis, statistical modelling, Gaussian Process regression, time-series analysis, signal detection, simulation, uncertainty analysis and data visualisation

Technology & Engineering: Sensor integration, data acquisition, hardware prototyping, circuit design, embedded systems and instrument calibration

Business & Consulting: Problem solving, stakeholder engagement, project management, data-driven decision-making, research, strategy and presentation

Software: Microsoft Excel, PowerPoint, Word, VS Code

EXTRACURRICULAR

Institute of Physics Ambassador, Institute of Physics **2026 - Present**

- Acting as a liaison between students and the professional physics community while promoting industry opportunities, events and career development.

Physics Society, Trinity College Dublin **2023 - Present**

Institute of Physics Representative. Formerly: Chair, Treasurer, Public Relations Officer

- Represent student interests in discussions with the Institute of Physics and promoted engagement with professional opportunities and events.
- Led the society committee, coordinated activities, and oversaw strategic planning and event delivery, managing finances and social media, publicity and outreach campaigns.

Deputy Science and Technology Editor, Trinity News **2025 - 2026**

- Contributed editorial and science communication work covering science and technology topics.

OTHER EXPERIENCES

Social Media Ambassador **2023 - Present**

- Collaborated with other ambassadors to develop ideas to improve and increase engagement across social media content. Managed and delivered projects from initial concept through to completion.

Social Media Manager **May 2023 – Nov 2024**

- Managed social media content for the Councillor John Walsh’s account. Created informative content while supporting John’s local election campaign. Designed graphics and social media materials and contributed to manifesto development and campaign strategy.

REFERENCES AVAILABLE ON REQUEST